10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 3.6 — Simultaneous Linear Equations

Unit 3 | Solved Exercise | New Book (PECTAA)

Exercise 3.6 (Solved)

1. Solve by matrix inversion method, if possible.

Write the two equations as AX = B, where A is the matrix of coefficients. Then X = $A^{-1}B$, and the solution is possible only when $|A| \neq 0$.

(i) 2x + 5y = 19, 4x − 3y = −1

Write the equations in matrix form AX = B

$$\begin{bmatrix} 2 & 5 \\ 4 & -3 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 19 \\ -1 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & 5 \\ 4 & -3 \end{vmatrix}$$

$= (2)(-3) - (5)(4)$

$= -26$

$$\text{adj } A = \begin{bmatrix} -3 & -5 \\ -4 & 2 \end{bmatrix}$$

So $A^{-1} = (1 \div |A|) \times \text{adj } A$

$$A^{-1} = \frac{-1}{26} \begin{bmatrix} -3 & -5 \\ -4 & 2 \end{bmatrix}$$

$$X = A^{-1}B = \frac{-1}{26} \begin{bmatrix} -3 & -5 \\ -4 & 2 \end{bmatrix} \times \begin{bmatrix} 19 \\ -1 \end{bmatrix}$$

$$X = \frac{-1}{26} \begin{bmatrix} -52 \\ -78 \end{bmatrix}$$

$$X = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Answer: x = 2, y = 3

(ii) 3x + 2y = 7, 5x − y = 16

Write the equations in matrix form AX = B

$$\begin{bmatrix} 3 & 2 \\ 5 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 16 \end{bmatrix}$$

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$$|A| = \begin{vmatrix} 3 & 2 \\ 5 & -1 \end{vmatrix}$$

$= (3)(-1) - (2)(5)$

$= -13$

$$\text{adj } A = \begin{bmatrix} -1 & -2 \\ -5 & 3 \end{bmatrix}$$

So $A^{-1} = (1 \div |A|) \times \text{adj } A$

$$A^{-1} = \frac{-1}{13} \begin{bmatrix} -1 & -2 \\ -5 & 3 \end{bmatrix}$$

$$X = A^{-1}B = \frac{-1}{13} \begin{bmatrix} -1 & -2 \\ -5 & 3 \end{bmatrix} \times \begin{bmatrix} 7 \\ 16 \end{bmatrix}$$

$$X = \frac{-1}{13} \begin{bmatrix} -39 \\ 13 \end{bmatrix}$$

$$X = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Answer: x = 3, y = −1

(iii) x − 2y = 9, 2x + 7y = −4

Write the equations in matrix form AX = B

$$\begin{bmatrix} 1 & -2 \\ 2 & 7 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ -4 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & -2 \\ 2 & 7 \end{vmatrix}$$

$= (1)(7) - (-2)(2)$

$= 11$

$$\text{adj } A = \begin{bmatrix} 7 & 2 \\ -2 & 1 \end{bmatrix}$$

So $A^{-1} = (1 \div |A|) \times \text{adj } A$

$$A^{-1} = \frac{1}{11} \begin{bmatrix} 7 & 2 \\ -2 & 1 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{11} \begin{bmatrix} 7 & 2 \\ -2 & 1 \end{bmatrix} \times \begin{bmatrix} 9 \\ -4 \end{bmatrix}$$

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$$X = \frac{1}{11}\begin{bmatrix} 55 \\ -22 \end{bmatrix}$$

$$X = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

Answer: x = 5, y = −2

(iv) 3x + 2y = 2, x − 2y = −2

Write the equations in matrix form AX = B

$$\begin{bmatrix} 3 & 2 \\ 1 & -2 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & 2 \\ 1 & -2 \end{vmatrix}$$

$= (3)(-2) - (2)(1)$
$= -8$

$$\text{adj } A = \begin{bmatrix} -2 & -2 \\ -1 & 3 \end{bmatrix}$$

So $A^{-1} = (1 \div |A|) \times \text{adj } A$

$$A^{-1} = \frac{-1}{8}\begin{bmatrix} -2 & -2 \\ -1 & 3 \end{bmatrix}$$

$$X = A^{-1}B = \frac{-1}{8}\begin{bmatrix} -2 & -2 \\ -1 & 3 \end{bmatrix} \times \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$X = \frac{-1}{8}\begin{bmatrix} 0 \\ -8 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Answer: x = 0, y = 1

2. Use Cramer's rule to solve the following pair of linear equations, if possible.

In Cramer's rule x = Dx ÷ D and y = Dy ÷ D. D is the determinant of the coefficients, Dx is made by replacing the x-column with the constants, and Dy by replacing the y-column with the constants.

(i) x + 4y = 4, 2x − y = 5

$$D = \begin{vmatrix} 1 & 4 \\ 2 & -1 \end{vmatrix}$$

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$= (1)(-1) - (4)(2) = -9$

$$D_x = \begin{vmatrix} 4 & 4 \\ 5 & -1 \end{vmatrix}$$

$= (4)(-1) - (4)(5) = -24$

$$D_y = \begin{vmatrix} 1 & 4 \\ 2 & 5 \end{vmatrix}$$

$= (1)(5) - (4)(2) = -3$

$$x = \frac{D_x}{D} = \frac{-24}{-9} = 8/3$$

$$y = \frac{D_y}{D} = \frac{-3}{-9} = 1/3$$

Answer: x = 8/3, y = 1/3

(ii) x + 2y = 7, 3x − 2y = −3

$$D = \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix}$$

$= (1)(-2) - (2)(3) = -8$

$$D_x = \begin{vmatrix} 7 & 2 \\ -3 & -2 \end{vmatrix}$$

$= (7)(-2) - (2)(-3) = -8$

$$D_y = \begin{vmatrix} 1 & 7 \\ 3 & -3 \end{vmatrix}$$

$= (1)(-3) - (7)(3) = -24$

$$x = \frac{D_x}{D} = \frac{-8}{-8} = 1$$

$$y = \frac{D_y}{D} = \frac{-24}{-8} = 3$$

Answer: x = 1, y = 3

(iii) 2x − 5y = −6, 4x − 3y = −12

$$D = \begin{vmatrix} 2 & -5 \\ 4 & -3 \end{vmatrix}$$

$= (2)(-3) - (-5)(4) = 14$

$$D_x = \begin{vmatrix} -6 & -5 \\ -12 & -3 \end{vmatrix}$$

$= (-6)(-3) - (-5)(-12) = -42$

$$D_y = \begin{vmatrix} 2 & -6 \\ 4 & -12 \end{vmatrix}$$

$= (2)(-12) - (-6)(4) = 0$

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$x = \frac{D_x}{D} = \frac{-42}{14} = -3$

$y = \frac{D_y}{D} = \frac{0}{14} = 0$

Answer: x = −3, y = 0

(iv) 3x + 2y = −1, 5x + 6y = 5

$D = \begin{vmatrix} 3 & 2 \\ 5 & 6 \end{vmatrix}$

$= (3)(6) - (2)(5) = 8$

$D_x = \begin{vmatrix} -1 & 2 \\ 5 & 6 \end{vmatrix}$

$= (-1)(6) - (2)(5) = -16$

$D_y = \begin{vmatrix} 3 & -1 \\ 5 & 5 \end{vmatrix}$

$= (3)(5) - (-1)(5) = 20$

$x = \frac{D_x}{D} = \frac{-16}{8} = -2$

$y = \frac{D_y}{D} = \frac{20}{8} = 5/2$

Answer: x = −2, y = 5/2

3. An electrical engineer wants to determine the current in the two branches A and B of a simple circuit. The system of equations is x + y = 7 and 2x − y = 12, where x is the current in branch A and y is the current in branch B. Find x and y by using matrices.

Solution

Let the current in branch A be x, and in branch B be y

$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 12 \end{bmatrix}$

$|A| = \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix}$

$= (1)(-1) - (1)(2)$

$= -1 - 2$

$= -3$

$\text{adj } A = \begin{bmatrix} -1 & -1 \\ -2 & 1 \end{bmatrix}$

$A^{-1} = \frac{1}{-3} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$

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$$X = A^{-1}B = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} 7 \\ 12 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 19 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 19/3 \\ 2/3 \end{bmatrix}$$

Answer: Current in branch A = 19/3, current in branch B = 2/3

4. Three forces act on a particle and must be in equilibrium, i.e. $F_1 + F_2 + F_3 = 0$, where F_1 = [[8], [x]], F_2 = [[−2], [−7]] and F_3 = [[y], [−1]]. Find the value of x and y.

Solution

$$\begin{bmatrix} 8 \\ x \end{bmatrix} + \begin{bmatrix} -2 \\ -7 \end{bmatrix} + \begin{bmatrix} y \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Add the corresponding elements of the three matrices

$$\begin{bmatrix} 6 + y \\ x - 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row: $6 + y = 0$

$y = -6$

From the second row: $x - 8 = 0$

$x = 8$

Answer: x = 8, y = −6

5. Two support beams A and B are holding up a combined load of 100 kN. Twice the load on beam A and three times the load on beam B equals 240 kN. Find the load of beam A and beam B by using matrices.

Solution

Let the load on beam A be x kN and on beam B be y kN

Combined load: $x + y = 100$

Twice A plus three times B: $2x + 3y = 240$

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 100 \\ 240 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}$$

$= (1)(3) - (1)(2)$

$= 1$

$$\text{adj } A = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{1}\begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$$

$$X = A^{-1}B = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix} \times \begin{bmatrix} 100 \\ 240 \end{bmatrix}$$

$$= \begin{bmatrix} 60 \\ 40 \end{bmatrix}$$

$$= \begin{bmatrix} 60 \\ 40 \end{bmatrix}$$

Answer: Load on beam A = 60 kN, load on beam B = 40 kN

6. In a 2D game world, two characters are moving along straight paths. One character moves along a line where the total of twice the horizontal position and the vertical position is 5, while the other moves along a line where the horizontal position is one more than the vertical position. Find their point of intersection by using matrices.

Solution

Let the horizontal position be x and the vertical position be y

First path: $2x + y = 5$

Second path: $x = y + 1$, i.e. $x - y = 1$

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix}$$

$= (2)(-1) - (1)(1)$

$= -3$

$$\text{adj } A = \begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \times \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Answer: The paths meet at (2, 1), i.e. x = 2 and y = 1

7. Two years ago a man was 5 times as old as his son was. After 6 years he will be 3 times as old as his son. Find their present ages by using matrices.

Solution

Let the present age of the man be x years, and of the son be y years

Two years ago: $x - 2 = 5(y - 2)$

$x - 2 = 5y - 10$

$x - 5y = -8$

After 6 years: $x + 6 = 3(y + 6)$

$x + 6 = 3y + 18$

$x - 3y = 12$

$$\begin{bmatrix} 1 & -5 \\ 1 & -3 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -8 \\ 12 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & -5 \\ 1 & -3 \end{vmatrix}$$

$= (1)(-3) - (-5)(1)$

$= 2$

$$\text{adj } A = \begin{bmatrix} -3 & 5 \\ -1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2}\begin{bmatrix} -3 & 5 \\ -1 & 1 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{2}\begin{bmatrix} -3 & 5 \\ -1 & 1 \end{bmatrix} \times \begin{bmatrix} -8 \\ 12 \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} 84 \\ 20 \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} 84 \\ 20 \end{bmatrix}$$

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$$= \begin{bmatrix} 42 \\ 10 \end{bmatrix}$$

Answer: Present age of the man = 42 years, present age of the son = 10 years

8. Two cyclists are 44 km apart and start out at the same time. If they go towards one another they meet in 2 hours, but if they go in the same direction the faster overtakes the slower in 7½ hours. Find their speeds by using matrices.

Solution

Let the speed of the faster cyclist be x km/h and of the slower be y km/h

Moving towards each other they cover 44 km in 2 hours

$2x + 2y = 44$, i.e. $x + y = 22$

Moving in the same direction the faster gains 44 km in 7½ hours

$(15/2)(x - y) = 44$

$x - y = 88/15$

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 22 \\ 88/15 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

$= (1)(-1) - (1)(1)$

$= -2$

$$\text{adj } A = \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} 22 \\ 88/15 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 22 + 88/15 \\ 22 - 88/15 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 418/15 \\ 242/15 \end{bmatrix}$$

$$= \begin{bmatrix} 209/15 \\ 121/15 \end{bmatrix}$$

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Answer: Speed of the faster cyclist = 209/15 km/h, speed of the slower = 121/15 km/h

9. The numerator of a fraction is 7 less than the denominator. If the numerator is increased by 3, the new fraction can be cancelled down to 3/4. Find the original fraction by using matrices.

Solution

Let the numerator be x and the denominator be y

The numerator is 7 less than the denominator

$x = y - 7$, i.e. $x - y = -7$

After increasing the numerator by 3, the fraction becomes 3/4

$(x + 3)/y = 3/4$

$4(x + 3) = 3y$

$4x + 12 = 3y$, i.e. $4x - 3y = -12$

$$\begin{bmatrix} 1 & -1 \\ 4 & -3 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -7 \\ -12 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & -1 \\ 4 & -3 \end{vmatrix}$$

$= (1)(-3) - (-1)(4) = -3 + 4$

$= 1$

$$\text{adj } A = \begin{bmatrix} -3 & 1 \\ -4 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} -3 & 1 \\ -4 & 1 \end{bmatrix}$$

$$X = A^{-1}B = \begin{bmatrix} -3 & 1 \\ -4 & 1 \end{bmatrix} \times \begin{bmatrix} -7 \\ -12 \end{bmatrix}$$

$$= \begin{bmatrix} 9 \\ 16 \end{bmatrix}$$

$$= \begin{bmatrix} 9 \\ 16 \end{bmatrix}$$

Answer: Numerator = 9, denominator = 16, so the fraction is 9/16

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